

GROWTH AND DEVELOPMENT



AS THE HUMAN BODY grows and develops, it follows a regular sequence of changes. After birth, a human being passes through infancy, childhood, puberty and adolescence, and into adulthood. The body grows at different rates at different times. Rapid growth, called a growth spurt, occurs during infancy and again at puberty, while growth is steady throughout childhood, but ceases in adulthood. In later life, the body ages as it becomes less efficient. Eventually, one or more of the body's systems stop working, and a person dies.

Changing proportions

Different parts of the body grow at different rates. Changing body proportions can be compared by fitting photographs of children and young adults into a panel that makes them appear the same height. The panel divides each body into eight equal parts. The head, for example, makes up one quarter of the height of a newborn baby, but only one eighth of the height of a 20-year-old.



2 months	2 years	4 years	7 years	12 years	20 years
55 cm	86 cm	110 cm	120 cm	145 cm	175 cm
(1 ft 10 in)	(2 ft 10 in)	(3 ft 8 in)	(4 ft)	(4 ft 10 in)	(5 ft 10 in)

From baby to child

During the first two years of life, a young human being grows and develops rapidly. A six-week-old baby is helpless and must have everything done for it, but by the age of two years, the baby can walk, talk, and feed itself. Growth and development is marked by a series of age milestones at which children have learned certain skills.

6 weeks

The baby sleeps when not being held or fed, and cries when distressed. She can follow objects with her eyes and listen to a person talking.



6 months

The baby can sit supported with her head up and back straight. She holds objects, squeals, and babbles.



10 months

The baby can crawl rapidly, pull herself up to a standing position, point to and pick up objects. She says her first words, usually "mama" and "dada".



8 months

The baby can sit up by herself, will try to crawl, and can stand if supported. She turns towards the sound of a familiar voice, and can imitate simple sounds.



Toddler can walk a few steps on her own.

Child needs help getting dressed but can put on own shoes and socks.

Child can dress and undress herself.



14 months

The child can stand alone and may walk without help. She speaks a few words, and tries to indicate what she wants.

2 years

She can run and jump, turn the pages of a book, identify pictures of familiar objects, and form a few short phrases.

4 years

The young girl now has good balance and can hop on one foot. She can draw simple pictures and copy some letters.

Developing bones

The skeleton is formed before birth from flexible cartilage. During childhood, this is replaced by bone, as revealed by X-rays. The skeleton continues to get larger and harder during the teenage years.



A newborn baby's skeleton is made of both bone and cartilage. Unlike bone, cartilage cannot be seen on an X-ray.



By the age of 6, the wrist bones are forming so that there are now more bones. Other bones have got harder and bigger.



By the age of 16, the 27 bones that make up the fingers, palm, and wrist, are now mature, hard, and adult in size and form.

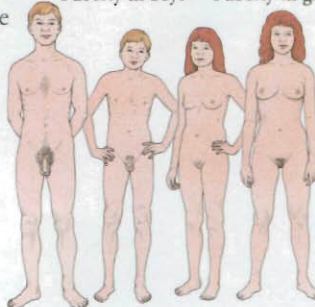
Adolescence and puberty

Adolescence is the whole process of growing up from a child to an adult. During adolescence, changes occur to a person's body and in the way they think and feel. Puberty is part of adolescence during which the body grows rapidly and changes shape, and boys and girls become sexually mature and able to reproduce.

Changes during puberty

In girls, puberty begins between the ages of 10–14. The body becomes rounder, breasts grow, and periods begin. In boys, puberty begins between the ages of 12–16. The body becomes more muscular, the testes produce sperm, and the voice deepens.

Puberty in boys Puberty in girls



Ageing

Growing old is a normal part of life. Humans age because the body's cells gradually become less efficient. Signs of ageing usually appear after 40 years of age. The body becomes less mobile, hair thins and turns grey, and the skin wrinkles. Bones become brittle and can break more easily. Exercise and a healthy diet can help to slow down the ageing process.



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